

Lecture 15: Game Theory

- **Origins: Nash and von Neumann**
- **Payoff matrices**
- **Nash Equilibrium**
- **Classic games**
- **The unique number game**

What is game theory?

- A framework for studying social interaction and decision-making
- Started in mathematics, then economics, but is used widely, e.g. in evolutionary biology

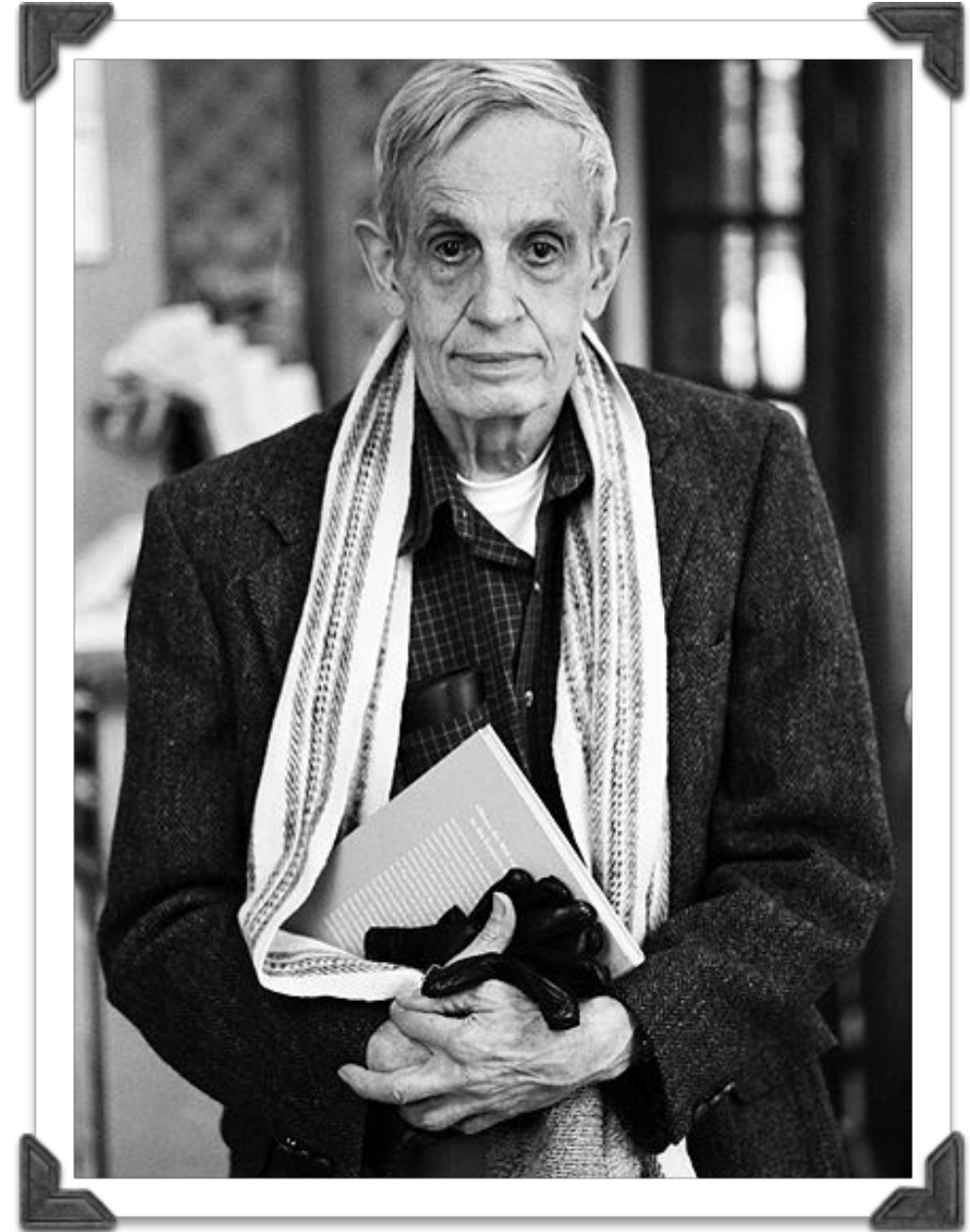


John Forbes Nash, Jr.

**PhD thesis, Princeton
(1950) 27 pages long**

Theses need not be long!

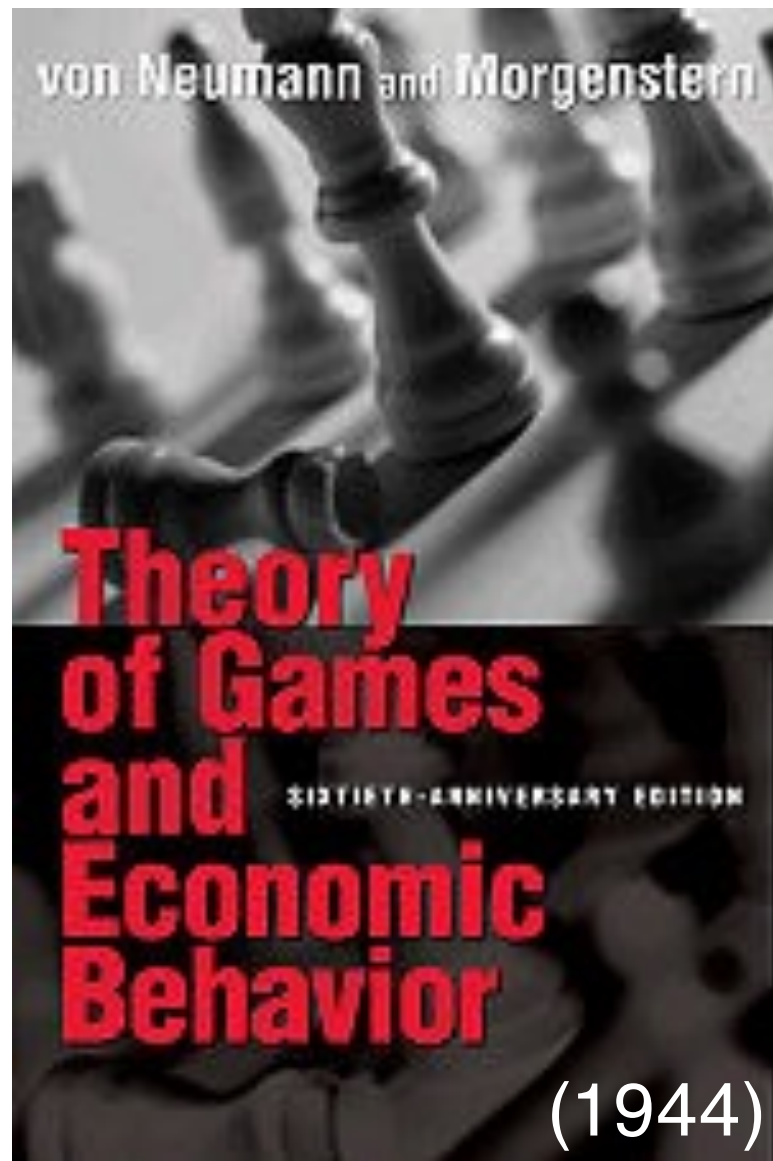
**[This is also true for senior
theses!]**



Kuhn's description of his thesis, at the awarding of Nash's Nobel prize in 1994

The entire thesis is 27 pages of typescript, very generously double-spaced. Frankly, I have always considered the most important sections to be the first 6 pages summarized above and the last pages (from page 21 to 26) on motivation, interpretation, and applications. For many years, I have accused John of padding the thesis in the middle section (15 pages in all).

Original Formulation of Game Theory



Nash's work built on von Neumann's original formulation.

His thesis was written as several papers—one very mathematical, the rest more explanatory

Two Player Game: my payoff

a_{ij}	You: Strategy 1	You: Strategy 2	
Me: Strategy 1	5	3	$a_{11} = 5$ $a_{21} = 3$
Me: Strategy 2	4	2	$a_{12} = 4$ $a_{22} = 2$

My payoff matrix: Gives me a reward, given the adoption of a strategy.

Other player has a payoffs given their strategies

Two Player Game: your payoff

b_{ij}	You: Strategy 1	You: Strategy 2
Me: Strategy 1	3	4
Me: Strategy 2	4	5

Your payoff matrix: Gives you a reward, given the adoption of a strategy.

Two Player Game

	You: Strategy 1	You: Strategy 2
Me: Strategy 1	$(a_{11}, b_{11}) = (5, 3)$	$(a_{12}, b_{12}) = (3, 4)$
Me: Strategy 2	$(a_{21}, b_{21}) = (4, 4)$	$(a_{22}, b_{22}) = (2, 5)$

Two Player Game

How should I play?

	You: Strategy 1	You: Strategy 2
Me: Strategy 1	$(a_{11}, b_{11}) = (5, 3)$	$(a_{12}, b_{12}) = (3, 4)$
Me: Strategy 2	$(a_{21}, b_{21}) = (4, 4)$	$(a_{22}, b_{22}) = (2, 5)$

Two Player Game

How should I play?

	You: Strategy 1	You: Strategy 2
Me: Strategy 1	$(a_{11}, b_{11}) = (5, 3)$	$(a_{12}, b_{12}) = (3, 4)$
Me: Strategy 2	$(a_{21}, b_{21}) = (4, 4)$	$(a_{22}, b_{22}) = (2, 5)$

- If you choose strategy 1, then I should choose strategy 1
- If you choose strategy 2, then I should choose strategy 1

Two Player Game

How should I play?

	You: Strategy 1	You: Strategy 2
Me: Strategy 1	$(a_{11}, b_{11}) = (5, 3)$	$(a_{12}, b_{12}) = (3, 4)$
Me: Strategy 2	$(a_{21}, b_{21}) = (4, 4)$	$(a_{22}, b_{22}) = (2, 5)$

For me, strategy 1 *dominates* over strategy 2

For me, strategy i *dominates* strategy j if $a_{ij} \geq a_{kj}$ for all j

Two Player Game

How should you play?

	You: Strategy 1	You: Strategy 2
Me: Strategy 1	$(a_{11}, b_{11}) = (5, 3)$	$(a_{12}, b_{12}) = (3, 4)$
Me: Strategy 2	$(a_{21}, b_{21}) = (4, 4)$	$(a_{22}, b_{22}) = (2, 5)$

For you, strategy 2 *dominates* over strategy 1

For you, strategy j *dominates* strategy l if $b_{ij} \geq b_{il}$ for all j

Dominant Strategy Equilibrium

	You: Strategy 1	You: Strategy 2
Me: Strategy 1	$(a_{11}, b_{11}) = (5, 3)$	$(a_{12}, b_{12}) = (3, 4)$
Me: Strategy 2	$(a_{21}, b_{21}) = (4, 4)$	$(a_{22}, b_{22}) = (2, 5)$

Both players play their dominant strategies.

Dominant Strategy Equilibrium

	You: Strategy 1	You: Strategy 2
Me: Strategy 1	$(a_{11}, b_{11}) = (5, 3)$	$(a_{12}, b_{12}) = (3, 4)$
Me: Strategy 2	$(a_{21}, b_{21}) = (4, 4)$	$(a_{22}, b_{22}) = (2, 5)$

Example of a **Nash Equilibrium**, each strategy is best for the respective player.

Neither player can improve by unilaterally changing strategy.

Another game

	You: Strategy 1	You: Strategy 2
Me: Strategy 1	(2, 1)	(0,0)
Me: Strategy 2	(0,0)	(1,1)

Does either player have a dominant strategy?

Another game

	You: Strategy 1	You: Strategy 2
Me: Strategy 1	(2 1)	(0,0)
Me: Strategy 2	(0,0)	(1 1)

No player has a dominant strategy.

Two Nash Equilibria. Neither player can improve payoff by unilaterally changing strategy.

Fundamental Questions

What should I do? What maximizes my payoff?

What strategy?

Pure: we play the same strategy every time we play

Mixed: we play one strategy with probability p , and a second strategy with probability $1 - p$

The Mathematical Paper

EQUILIBRIUM POINTS IN N -PERSON GAMES

BY JOHN F. NASH, JR.*

PRINCETON UNIVERSITY

Communicated by S. Lefschetz, November 16, 1949

One may define a concept of an n -person game in which each player has a finite set of pure strategies and in which a definite set of payments to the n players corresponds to each n -tuple of pure strategies, one strategy being taken for each player. For mixed strategies, which are probability distributions over the pure strategies, the pay-off functions are the expectations of the players, thus becoming polylinear forms in the probabilities with which the various players play their various pure strategies.

Any n -tuple of strategies, one for each player, may be regarded as a point in the product space obtained by multiplying the n strategy spaces of the players. One such n -tuple counters another if the strategy of each player in the countering n -tuple yields the highest obtainable expectation for its player against the $n - 1$ strategies of the other players in the countered n -tuple. A self-countering n -tuple is called an equilibrium point.

Note: This is basically the whole paper!

It relies on the notion of “fixed point theorems” (from analysis)

Less Mathematical Papers

THE BARGAINING PROBLEM¹

BY JOHN F. NASH, JR.

A new treatment is presented of a classical economic problem, one which occurs in many forms, as bargaining, bilateral monopoly, etc. It may also be regarded as a nonzero-sum two-person game. In this treatment a few general assumptions are made concerning the behavior of a single individual and of a group of two individuals in certain economic environments. From these, the solution (in the sense of this paper) of the classical problem may be obtained. In the terms of game theory, values are found for the game.

“Values are found for the game”

A TWO-PERSON bargaining situation involves two individuals who have the opportunity to collaborate for mutual benefit in more than one way. In the simpler case, which is the one considered in this paper, no action taken by one of the individuals without the consent of the other can affect the well-being of the other one.

TWO PERSON THEORY

In *Theory of Games and Economic Behavior* a theory of n -person games is developed which includes as a special case the two-person bargaining problem. But the theory there developed makes no attempt to find a value for a given n -person game, that is, to determine what it is worth to each player to have the opportunity to engage in the game. This determination is accomplished only in the case of the two-person zero sum game.

It is our viewpoint that these n -person games should have values; that is, there should be a set of numbers which depend continuously upon the set of quantities comprising the mathematical description of the game and which express the utility to each player of the opportunity to engage in the game.

Nash's Bargaining Example

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JOHN F. NASH, JR.

EXAMPLES

Let us suppose that two intelligent individuals, Bill and Jack, are in a position where they may barter goods but have no money with which to facilitate exchange. Further, let us assume for simplicity that the utility to either individual of a portion of the total number of goods involved is the sum of the utilities to him of the individual goods in that portion. We give below a table of goods possessed by each individual with the utility of each to each individual. The utility functions used for the two individuals are, of course, to be regarded as arbitrary.

Example:

<i>Bill's goods</i>	<i>Utility to Bill</i>	<i>Utility to Jack</i>
book	2	4
whip	2	2
ball	2	1
bat	2	2
box	4	1
<i>Jack's goods</i>		
pen	10	1
toy	4	1
knife	6	2
hat	2	2

Example:

<i>Bill's goods</i>	<i>Utility to Bill</i>	<i>Utility to Jack</i>
book	2	4
whip	2	2
ball	2	1
bat	2	2
box	4	1
<i>Jack's goods</i>		
pen	10	1
toy	4	1
knife	6	2
hat	2	2

Kuhn:

If it is clear that Nash had not read those writers, it is equally clear that this paper was written by a teenager. The evidence is that the objects in the example to be bargained over are a bat, a ball, a toy, and a knife. No amount of urging by his colleagues, or by the editor of *Econometrica*, persuaded John to change this example.

Another game

	You: Strategy 1	You: Strategy 2
Me: Strategy 1	$(2, 1)$	$(1, 2)$
Me: Strategy 2	$(1, 2)$	$(2, 1)$

Another game

	You: Strategy 1	You: Strategy 2
Me: Strategy 1	(2, 1)	(1, 2)
Me: Strategy 2	(1, 2)	(2, 1)

If you choose strategy 1, then I choose strategy 1
If you choose strategy 2, then I choose strategy 2

Another game

	You: Strategy 1	You: Strategy 2
Me: Strategy 1	(2, 1)	(1, 2)
Me: Strategy 2	(1, 2)	(2, 1)

If I choose strategy 1, then you choose strategy 2
If I choose strategy 2, then you choose strategy 1

Another game

	You: Strategy 1	You: Strategy 2
Me: Strategy 1	$(\textcircled{2}, 1)$	$(1, \boxed{2})$
Me: Strategy 2	$(1, \boxed{2})$	$(\textcircled{2}, 1)$

No Nash Equilibria

Nash's Main Theorem

There is always a Nash Equilibrium using mixed strategies!

He explains his proof in words in his short paper (1 page).

In equations, payoff for strategy p with payoff matrix A against strategy q is

$$\mathbf{p} \cdot \mathbf{A} \mathbf{q} = \sum_{i=1}^n \sum_{j=1}^n p_i A_{ij} q_j$$

For any A , there is at least one strategy q such that for all opposing strategies p

$$\mathbf{q} \cdot \mathbf{A} \mathbf{q} \geq \mathbf{p} \cdot \mathbf{A} \mathbf{q}$$

Here q is a Nash equilibrium.

**Do such games appear in
real life?**

Bertrand Russell, 1959

Since the nuclear stalemate became apparent, the Governments of East and West have adopted the policy which Mr. Dulles calls 'brinkmanship'. This is a policy adapted from a sport which, I am told, is practised by some youthful degenerates. This sport is called 'Chicken!'. It is played by choosing a long straight road with a white line down the middle and starting two very fast cars towards each other from opposite ends. Each car is expected to keep the wheels of one side on the white line. As they approach each other, mutual destruction becomes more and more imminent. If one of them swerves from the white line before the other, the other, as he passes, shouts 'Chicken!', and the one who has swerved becomes an object of contempt. As played by irresponsible boys, this game is considered decadent and immoral, though only the lives of the players are risked. But when the game is played by eminent statesmen, who risk not only their own lives but those of many hundreds of millions of human beings, it is thought on both sides that the statesmen on one side are displaying a high degree of wisdom and courage, and only the statesmen on the other side are reprehensible. This, of course, is absurd. Both are to blame for playing such an incredibly dangerous game. The game may be played without misfortune a few times, but sooner or later it will come to be felt that loss of face is more dreadful than nuclear annihilation. The moment will come when neither side can face the derisive cry of 'Chicken!' from the other side. When that moment is come, the statesmen of both sides will plunge the world into destruction.[3]

The Chicken Game



Cars drive towards each other at high speeds on a single lane road.



Do I turn off the road, or keep going?

**My
payoff
matrix**

	Go for it	Chicken
Go for it	-c	b
Chicken	-b	0

-c = cost of crashing, $c > 0$

b = gain from winning, $c > b > 0$

The Chicken Game

**My
payoff
matrix**

	Go for it	Chicken
Go for it	-c	b
Chicken	-b	0

-c = cost of crashing, $c > 0$

b = gain from winning, $c > b > 0$

In a Nash equilibrium, neither player can *unilaterally* improve its payoff by changing strategies.

Are there Nash equilibria?

Discuss

The Chicken Game

Full
payoff
matrix

	Go for it	Chicken
Go for it	$(-c, -c)$	$(b, -b)$
Chicken	$(-b, b)$	$(0, 0)$

$-c$ = cost of crashing, $c > 0$

b = gain from winning, $c > b > 0$

In a Nash equilibrium, neither player can *unilaterally* improve its payoff by changing strategies.

Are there Nash equilibria?

YES: either (go for it, chicken) or (chicken, go for it)

The Prisoner's Dilemma



- **Two prisoners committed a crime together. They are separated and cannot talk to each other.**
- **The prosecutor offers each the same deal: confess, become a witness to the prosecution, and no jail time.**
- **If one prisoner confesses while other does not, the first goes free and the second goes to jail for 10 years.**
- **If both confess they will each go to jail for 7 years.**
- **If neither confess they go to jail for 1 year (since nothing can be proven).**

Discovered by Al Tucker, Nash's PhD advisor

HARSANYI

When did Tucker discover the Prisoner's Dilemma?

KUHN

Al Tucker was on leave at Stanford in the Spring of 1950 and, because of the shortage of offices, he was housed in the Psychology Department. One day a psychologist knocked on his door and asked what he was doing. Tucker replied: "I'm working on game theory.", and the psychologist asked if he would give a seminar on his work. For that seminar, Al Tucker invented the Prisoner's Dilemma as an example of game theory, Nash equilibria, and the attendant paradoxes of non-socially-desirable equilibria. A truly seminal example, it inspired dozens of research papers and several entire books.

The Prisoner's Dilemma



	Remain Silent	Confess
Remain Silent	$(-1, -1)$	$(-10, 0)$
Confess	$(0, -10)$	$(-7, -7)$

What should they do?

Discuss

The unique number game

- **Take n players**
- **They each choose an integer from 0 to 99**
- **The person who chose the lowest unique number wins**

Two example games with five players

Player	A	B	C	D	E
Choice	0	4	3	2	0

Player D wins

Player	A	B	C	D	E
Choice	0	3	3	0	0

No unique number – draw

Let's play!

- **We'll give you five minutes to think and discuss general strategy**
- **We want you to make a choice in secret, without cooperation with your neighbors**
- **Write your choice and your name on the paper**
- **We'll collect all the guesses and determine the winner**
- **The winner will be rewarded with chocolate**